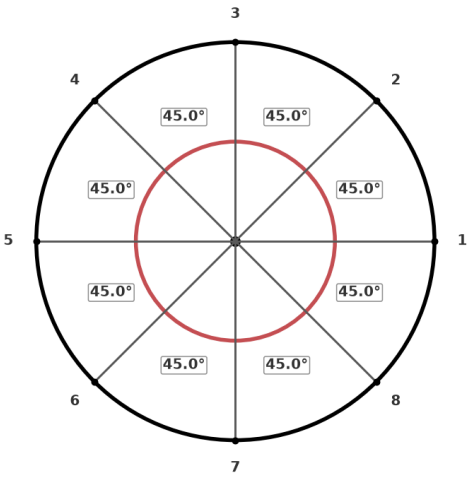
Shifting an inner circle: angular dipole on its surface

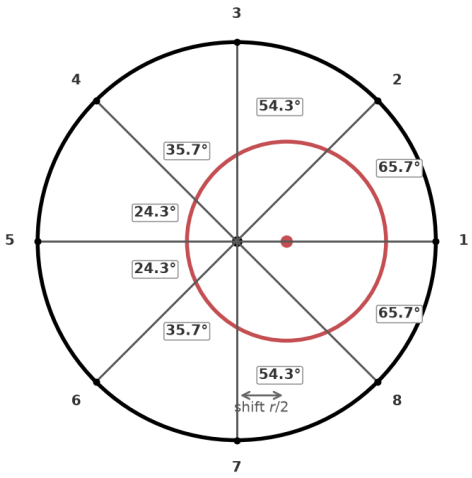
Simple geometry only. Outer circle radius $R = 2r$; inner circle radius r shifted by $r/2$. Guides show how equal chunks of the outer circle map onto the inner surface. No cosmology labels.

Shifting the inner circle changes the angular pattern on its surface ($R = 2r$, shift $= r/2$).

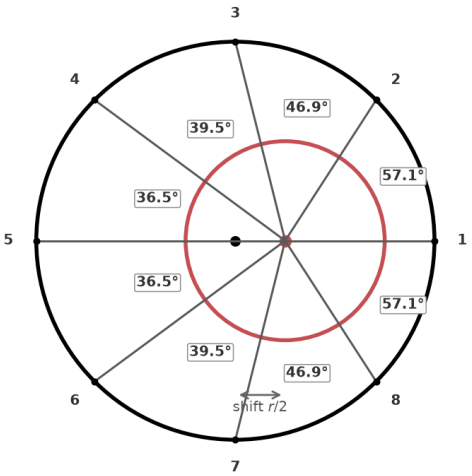
0 — concentric



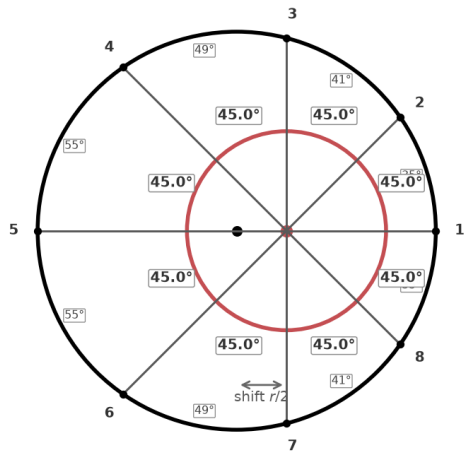
1 — inner shifted; guides to outer center



2 — equal outer arcs; guides to inner center



4 — equal 45° at inner center; unequal outer arcs



Scenarios

- **0** Concentric: equal angles all around (all 45°).
- **1** Inner shifted; guides still through the outer center: angles on the inner surface are unequal.
- **2** Equal outer arcs, guides to the inner center: unequal angles on the inner surface.
- **4** Equal 45° at the inner center: each wedge takes a different outer arc length.

Shifting the inner circle produces an angular dipole pattern on its surface (crowding on the side toward the outer center).